



Lecture 3 Assignment

Question 1

(a)

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix}; A^2 = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 2 \\ -4 & -2 & 1 \end{bmatrix}; A^3 = \begin{bmatrix} -4 & -2 & 1 \\ 4 & -3 & 2 \\ -3 & -4 & -4 \end{bmatrix}$$

(b) $A^3 = A^2 + A - 5I$

$$A^3 = \begin{bmatrix} -4 & -2 & 1 \\ 4 & -3 & 2 \\ -3 & -4 & -4 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 2 \\ -4 & -2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c)

$$(i) A^4 = AA^3 = A(A^2 + A - 5I) = A^3 + A^2 - 5A$$

$$A^4 = (A^2 + A - 5I) + A^2 - 5A = 2A^2 - 4A - 5I$$

$$(ii) A^3 = A^2 + A - 5I$$

$$\Rightarrow 5I = A^2 + A - A^3$$

$$\Rightarrow I = \frac{1}{5}(A^2 + A - A^3)$$

$$\Rightarrow A^{-1} = \frac{1}{5}(A + I - A^2)$$

Question 2

Let $X = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, and given $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

So, $PX = XP$ (Commutative Property)

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a_{11} & a_{12} \\ 0 & a_{21} & a_{22} \\ 0 & a_{31} & a_{32} \end{bmatrix}$$

Equating the matrices, we have $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{11} & a_{12} \\ 0 & 0 & a_{11} \end{bmatrix}$ which is can be considered

as $\begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$.

Question 3

- (a) $(A + A^T)$ is symmetric
 $(A + A^T)^T = A^T + (A^T)^T \quad \because (A + B)^T = A^T + B^T$
 $(A + A^T)^T = A^T + A = A + A^T \quad \because (A^T)^T = A, \quad (A + B) = (B + A)$
Hence $(A + A^T)$ is symmetric matrix.
- $(A - A^T)$ is skew-symmetric
 $(A - A^T)^T = A^T - (A^T)^T \quad \because (A + B)^T = A^T + B^T$
 $(A - A^T)^T = A^T - A = -(A - A^T) \quad \because (A^T)^T = A, \quad (A + B) = (B + A)$
Hence $(A - A^T)$ is skew-symmetric matrix.
- (b) AA^T and $A^T A$ are symmetric.
 $(AA^T)^T = (A^T)^T A^T = AA^T \quad \because (AB)^T = B^T A^T, \quad (A^T)^T = A$
 $(A^T A)^T = A^T (A^T)^T = A^T A$
- (c) $A^2 = A$ and A^{-1} exists then $A = I$.
 $A^{-1}A^2 = A^{-1}A$
 $A = I$.
 $A^2 A^{-1} = AA^{-1}$
 $A = I$.
- (d) $(A^{-1})^T = (A^T)^{-1}$.
We know $I = I$.
 $(A^{-1}A) = I. \quad \because (A^{-1}A) = I$
Apply transpose on both side
 $(A^{-1}A)^T = I^T$.
 $A^T(A^{-1})^T = I. \quad \because (AB)^T = B^T A^T, \quad I^T = I$
 $A^T(A^{-1})^T = I$.
 $(A^T)^{-1}A^T(A^{-1})^T = (A^T)^{-1}I$.
 $I(A^{-1})^T = (A^T)^{-1}$.
 $(A^{-1})^T = (A^T)^{-1}$

Question 5

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \quad X = \begin{bmatrix} x & y \\ z & t \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

So, we take $AX = XB$

$$\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x & y \\ z & t \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} x & y \\ x + 2z & y + 2t \end{bmatrix} = \begin{bmatrix} 2x - y & -x + 2y \\ 2z - t & -z + 2t \end{bmatrix}$$

By equating the matrices we have

$$x = y, x = -t, y = -z$$

$$\text{So, the } X \text{ matrix will be } \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x & x \\ -x & -x \end{bmatrix} = x \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

Question 6

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \text{ (3x3 Matrix) and } A^T = -A \text{ (Skew-symmetric) so}$$

$$\begin{aligned}
\begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} &= - \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \\
\begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \\
\begin{bmatrix} 2a_{11} & a_{21} + a_{12} & a_{31} + a_{13} \\ a_{12} + a_{21} & 2a_{22} & a_{32} + a_{23} \\ a_{13} + a_{31} & a_{23} + a_{32} & 2a_{33} \end{bmatrix} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.
\end{aligned}$$

Equating the components of the matrices, so we have diagonal entries a_{11}, a_{22}, a_{33} equals to 0.

Question 7

Let's take $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $B = \begin{bmatrix} x & y \\ z & t \end{bmatrix}$ and we need to show $(AB)^T = B^T A^T$.

$$\begin{aligned}
\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & y \\ z & t \end{bmatrix} \right)^T &= \begin{bmatrix} x & y \\ z & t \end{bmatrix}^T \begin{bmatrix} a & b \\ c & d \end{bmatrix}^T \\
\left(\begin{bmatrix} ax + bz & ay + bt \\ cx + dz & cy + dt \end{bmatrix} \right)^T &= \begin{bmatrix} x & z \\ y & t \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} \\
\begin{bmatrix} ax + bz & cx + dz \\ ay + bt & cy + dt \end{bmatrix} &= \begin{bmatrix} ax + bz & cx + dz \\ ay + bt & cy + dt \end{bmatrix}
\end{aligned}$$